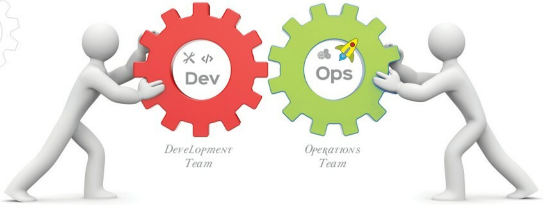
Starting the DevOps Journey Using Cucumber and Selenium



DevOps is a software building process which emphasises communication and collaboration between the teams involved in product management, software development, and operations. Cucumber is a behaviour driven development tool, which when combined with Selenium, a test recording and playback tool, improves the development team's efficiency and productivity.

It is often said that continuous change is the law of the universe and the same is true in the software industry. We have seen a variety of software development models, starting from Waterfall, V and spiral models to the incremental option. All these models have different requirements and guidelines and suit different scenarios. These days, most organisations have embraced the Agile methodology for software development.

The Agile method of developing software and applications focuses on delivering high quality products frequently and consistently, which leads to an increase in business value and profits.

Table 1 lists the differences between the Waterfall and Agile software development approaches.

Table 1

Waterfall model	Agile model
Processes are divided into different phases such as design, development, tests, etc.	Here the software development process is divided into sprints, usually spanning a few weeks.
First, the final product to be developed is defined according to the customers' needs and then the different phases begin until a 'finished' product is released.	Small targets are finalised and work is done on them, sprint by sprint. This means the final product is developed in small iterations.
It's difficult to incorporate changes in the requirements since it involves cost and time.	Due to the iterative approach being used in this model, it becomes easy to incorporate changes in the requirements.
Client participation in the development becomes negligible in this process.	Clients or product owners are actively involved in the process and constant feedback is given.

You can see that the Waterfall model can cause overshooting in time and resources, which can lead to huge losses to the company in terms of profits and user satisfaction. To avoid this, organisations have started adopting the Agile model. There are other reasons too, for choosing this model, some of which are listed below.

- *Client engagement:* In the Agile model, the client is engaged in the software development process at every step — before, during and after the sprint. This helps the development team to understand the client's vision clearly so that defect-free and high quality software can be developed and delivered in less time.
- *Transparency:* Since the client is actively involved in all the sprint activities, ranging from feature prioritisation and planning to reviewing and, finally, to deployment, this